Stress Analysis on an Inclined Plane

Problem

M3:A bar with a diameter of 25 mm is subjected to a direct pull of 60 kN. Determine the normal and shear stresses on a plane inclined at 30° to the axis.

Solution

Given data:

- Diameter of the bar, $d = 25\,\text{mm}$
- Axial force, $P = 60\,\text{kN} = 60 \times 10^3\,\text{N}$
- Inclination angle, $\theta = 30^\circ$

Calculate the cross-sectional area of the bar

The cross-sectional area of the bar is given by:

$$A = \frac{\pi d^2}{4}$$

Substituting $d = 25\,\text{mm} = 0.025\,\text{m}$ **(1 mark)**:

$$A = \frac{\pi (0.025)^2}{4} = 4.91 \times 10^{-4}\,\text{m}^2$$

Calculate the direct stress

The normal stress σ is given by:

$$\sigma = \frac{P}{A}$$

Substituting $P = 60 \times 10^3\,\mathrm{N}$ and $A = 4.91 \times 10^{-4}\,\mathrm{m}^2$: **(1 mark)**

$$\sigma = \frac{60 \times 10^3}{4.91 \times 10^{-4}} = 122.20\,\mathrm{MPa}$$

Resolve stresses on the inclined plane

The normal stress σ_θ and shear stress τ_θ on a plane inclined at an angle θ are given by:

$$\sigma_\theta = \sigma \cos^2\theta$$

$$\tau_\theta = \frac{1}{2}\sigma \sin(2\theta)$$

Substituting $\sigma = 122.20\,\mathrm{MPa}$ and $\theta = 30^\circ$:

For σ_θ **(1.5 marks)**:

$$\sigma_\theta = 122.20 \cos^2(30^\circ) = 122.20 \times \frac{3}{4} = 91.65\,\mathrm{MPa}$$

For τ_θ **(1.5 marks)**:

$$\tau_\theta = 0.5 \times 122.20 \times \sin(60^\circ) = 0.5 \times 122.20 \times \frac{\sqrt{3}}{2} = 52.91\,\mathrm{MPa}$$

Final Results

- Normal stress on the inclined plane: $91.65\,\mathrm{MPa}$
- Shear stress on the inclined plane: $52.91\,\mathrm{MPa}$